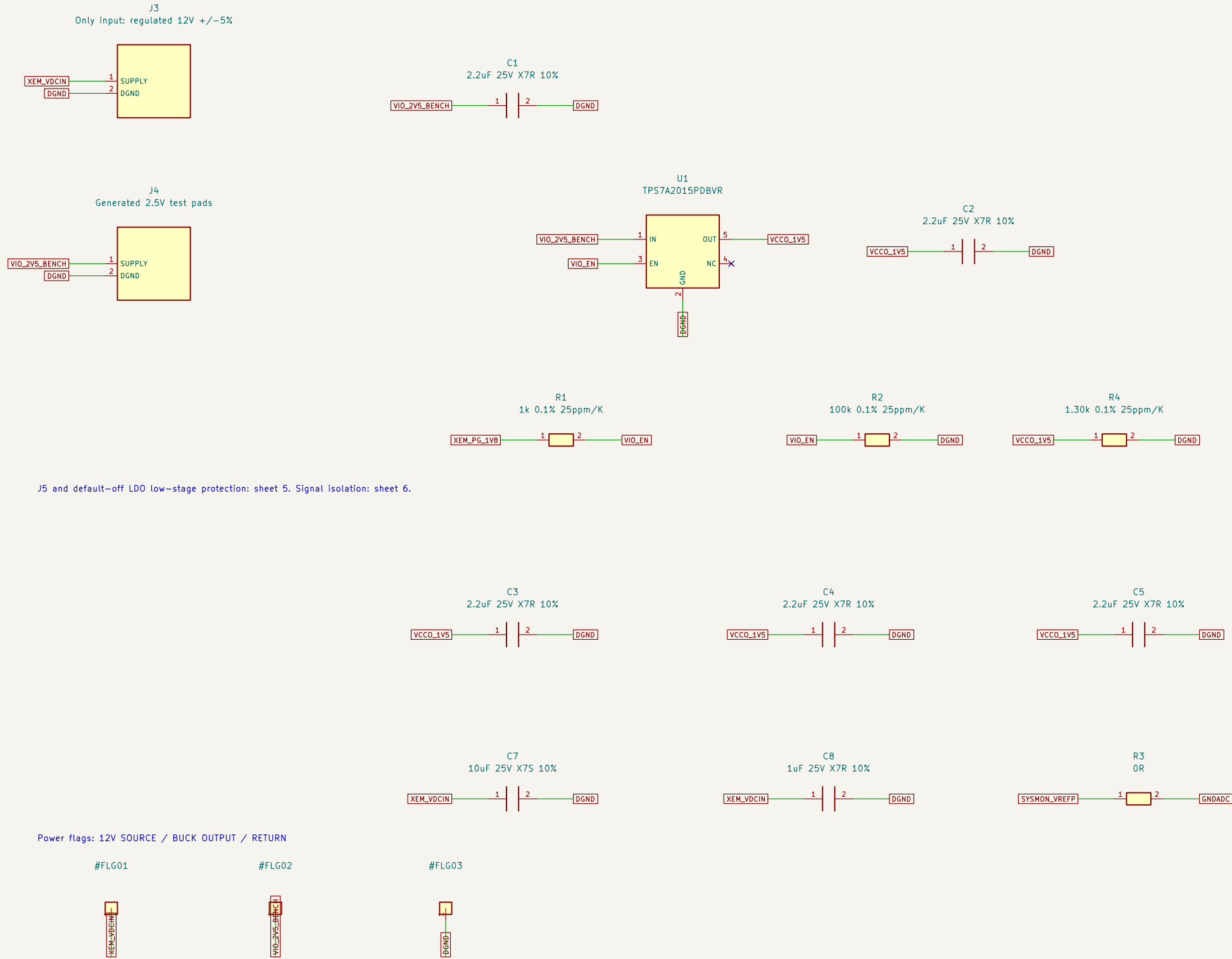


Confirmed plan: reuse Gerald LDO + existing ASIC; rigid adapter directly to XEM8305.

This board routes data and generates FPGA VCCO. Gerald board retains ASIC regulators.

No micro-HDMI interface is required for this backup path.



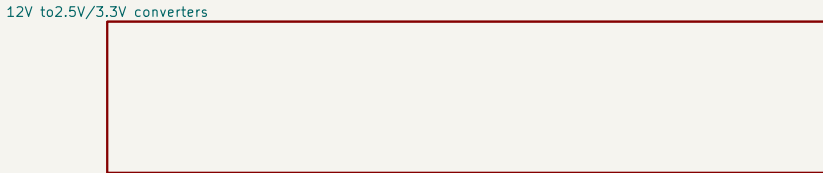
C3/C4/C5/C6: place at bank64/84/85/86 power contacts; preserve effective >=1uF.

Bring-up order: ASIC and stimulation supplies OFF -> XEM -> stable1.5V VCCO -> safe GPIO -> ASIC.

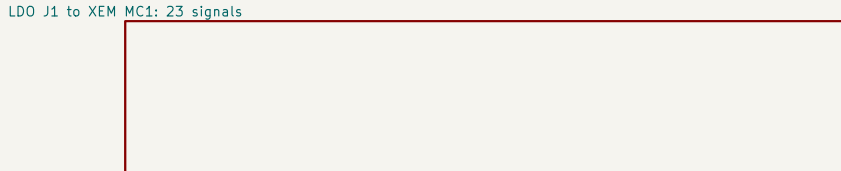
Power down ASIC before XEM. Default-off TMUX links and supervised low-stage feed are on sheets 5/6.

Existing high-voltage supply and AGND arrangement must be reviewed with Gerald; not invented here.

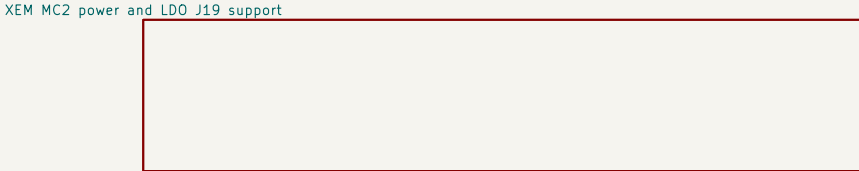
VIO planning envelope: 0-150 mA external aggregate plusbleeder at <=55 C ambient; physical load/thermal proof open.



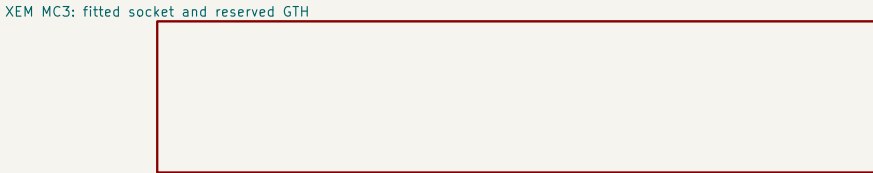
File: 07_Single_12V_Input.kicad_sch



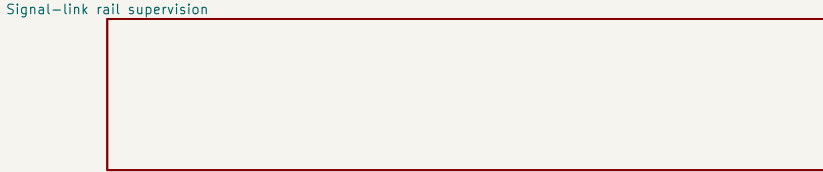
File: 01_LDO_And_MC1.kicad_sch



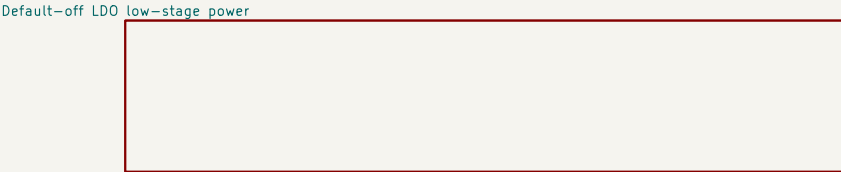
File: 02_MC2_And_Mechanical_J19.kicad_sch



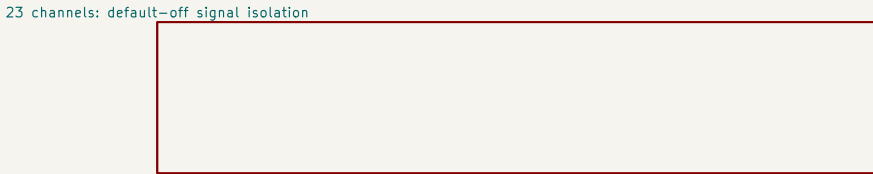
File: 03_MC3_Reserved_High_Speed.kicad_sch



File: 06_Link_Supervision.kicad_sch



File: 04_Protected_LDO_Feed.kicad_sch



File: 05_Signal_Isolation.kicad_sch

Bench candidate: startup / timing / mechanics unverified

FPGA PCB project

Sheet: /

File: LDO_to_XEM8305_Direct_R3.kicad_sch

Title: LDO to XEM8305: bench power

Size: A2

Date: 2026-10-05

Rev: R3

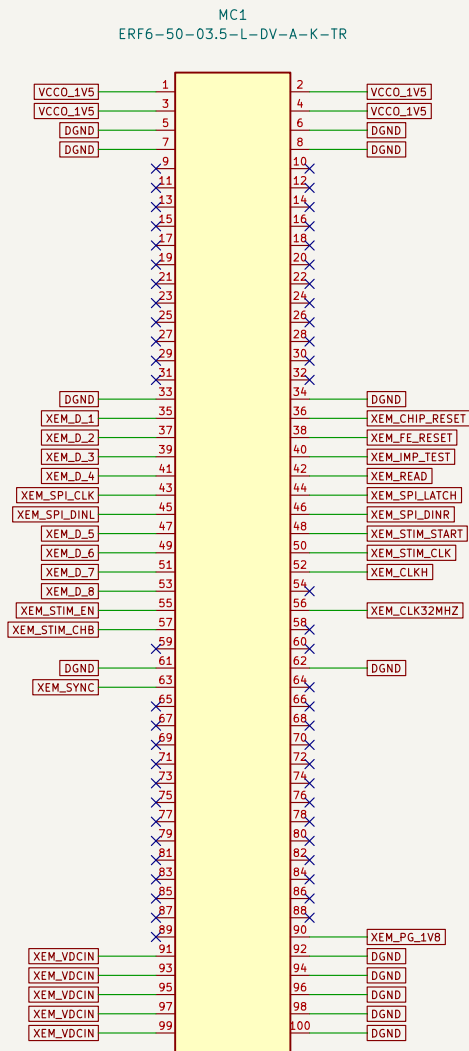
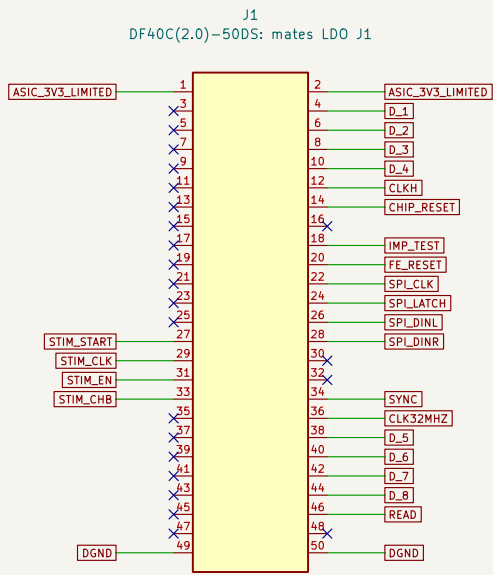
KiCad E.D.A. 10.0.6

Id: 1/8

Howard confirmed existing J1 + J19 on 2 October; J2 remains ASIC-facing.

23 signal routes preserved. FPGA pin/bank assignment is a proposal verified against official CSV.

Digital 1.5 V limits, direction, timing and reset/stim inactive levels still require confirmation.



Existing J1 pins47/48 have only an isolated VCC pair; adapter intentionally NC.

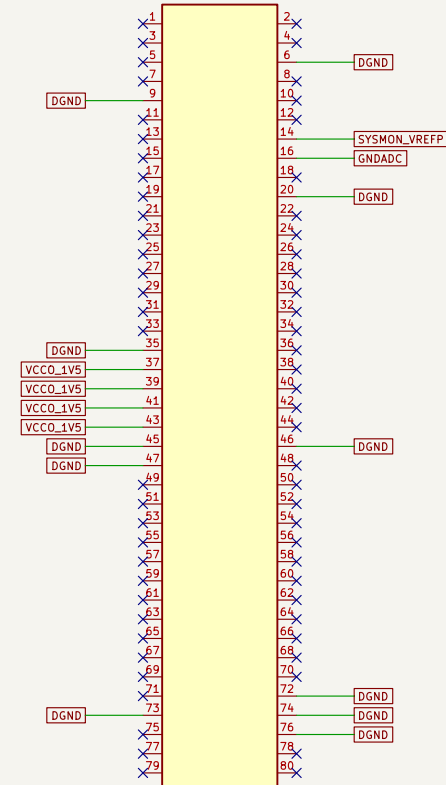
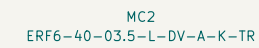
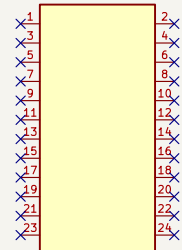
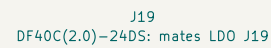
All assigned signal balls use Bank64. CLKH/SYNC/CLK32MHZ use clock-capable sites.

No claim that source4cmPCB experience validates this longer route or eight-chip system.

TMUX1511 switches isolate these nominal1.5V paths; no voltage translation is performed.

J19 is a confirmed mating connector; every contact remains electrically unused.

Do not assign power to J19 without a revised Gerald PCB pin map.



All three expansion sockets are fitted. MC3 high-speed signals are explicitly reserved.

All four external VCCO banks share the proposed 1.5V rail. Verify actual bank currents.

Module1.8V is an output, not a3.3V ASIC source. No tie to bench3.3V or other outputs.

The adapter is a separate board. Do not edit or rewire the supplied LDO project.

Bench candidate: startup / timing / mechanics unverified

FPGA PCB project

Sheet: /XEM MC2 power and LDO J19 support/

File: 02_MC2_And_Mechanical_J19.kicad_sch

Title: XEM MC2 power and LDO J19 support

Size: A3

Date: 2026-10-05

Rev: R3

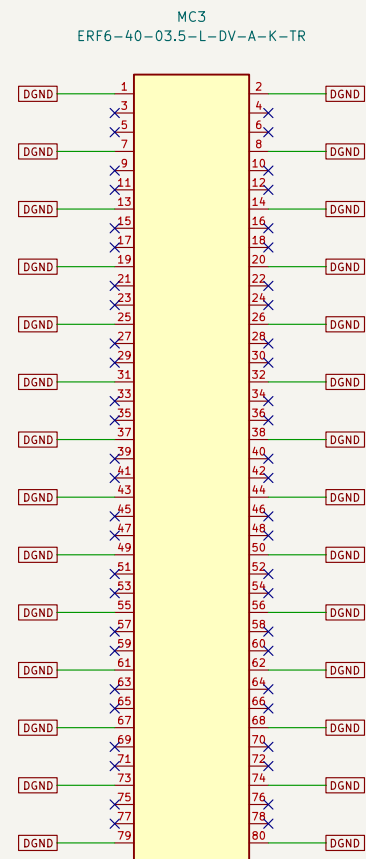
KiCad E.D.A. 10.0.6

Id: 3/8

MC3 physically mates to the XEM8305: 80 contacts, 5.0 mm mating gap.

28 DGND contacts connect to the reference planes; 52 GTH/REFCLK contacts remain NC.

GTH TX/RX and reference clocks are dedicated high-speed pins, not ASIC LVCMOS15 GPIO.



A defined high-speed endpoint, impedance and coupling design are required before signal use.

Bench candidate: startup / timing / mechanics unverified

FPGA PCB project

Sheet: /XEM MC3: fitted socket and reserved GTH/

File: 03_MC3_Reserved_High_Speed.kicad_sch

Title: XEM MC3: fitted socket and reserved GTH

Size: A3

Date: 2026-10-05

Rev: R3

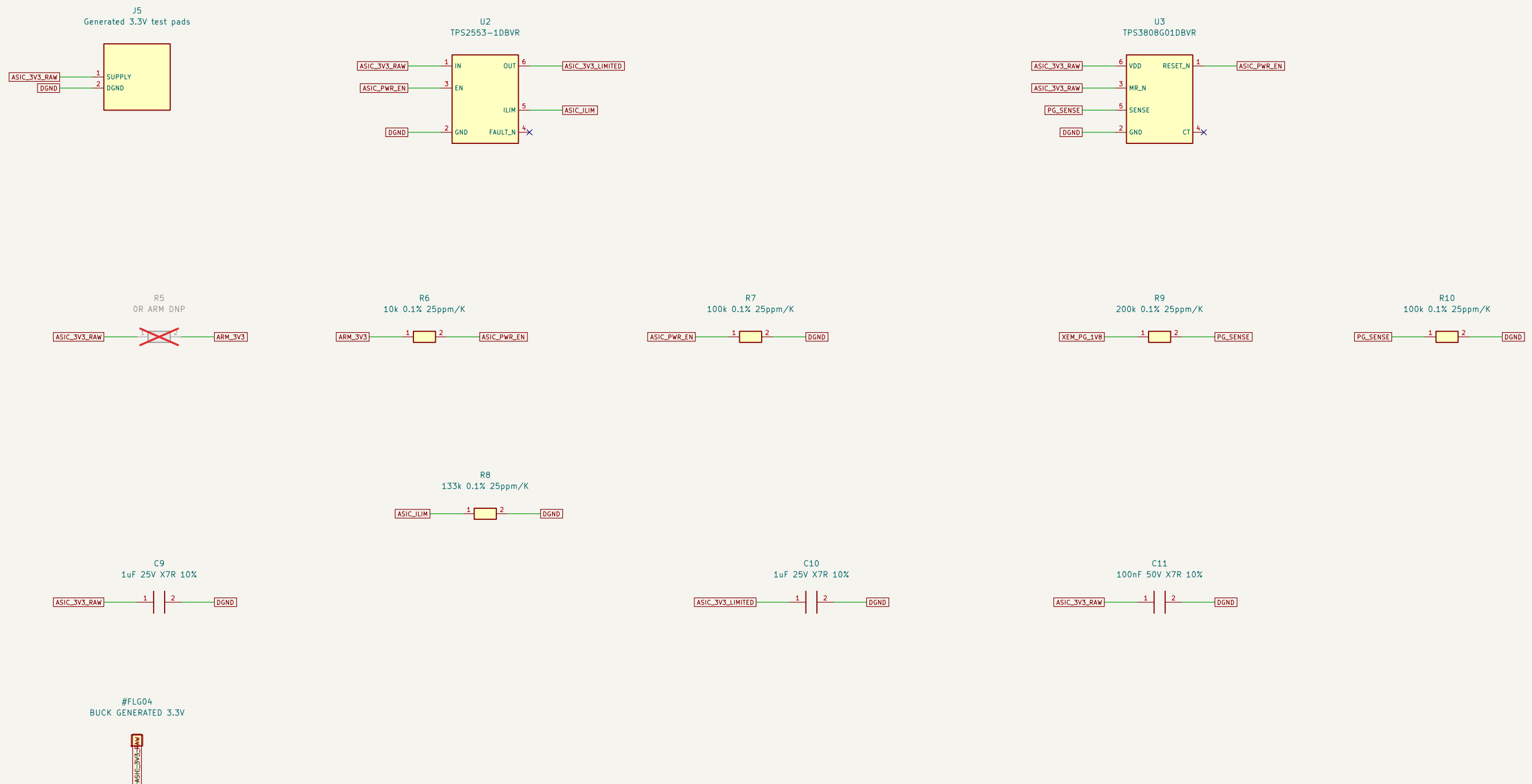
KiCad E.D.A. 10.0.6

Id: 4/8

R5 ARM link is DNP for shipping. Populate only after unpowered seating and quiet-GPIO checks.

Onboard low-stage rail: nominal 3.3V; ≤ 150 mA continuous / ≤ 120 mA startup planning load.

This feed does not generate or qualify the original stimulation high-voltage inputs or AGND bond.



FAULT_N and CT are intentionally NC. Fault recovery: remove ARM and resolve fault before rearming.

Check startup inrush on the actual LDO/ASIC; limiter may latch if load/capacitance exceed the planning case.

3.3V buck must satisfy worst switch/copper drop and $\geq 3.133\text{V}$ LINK supervisor release at the load node.

Bench candidate: startup / timing / mechanics unverified

FPGA PCB project

Sheet: /Default-off LDO low-stage power/

File: 04_Protected_LDO_Feed.kicad_sch

Title: Default-off LDO low-stage power

Size: A3

Date: 2026-10-05

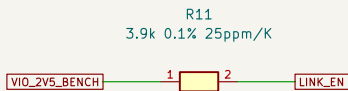
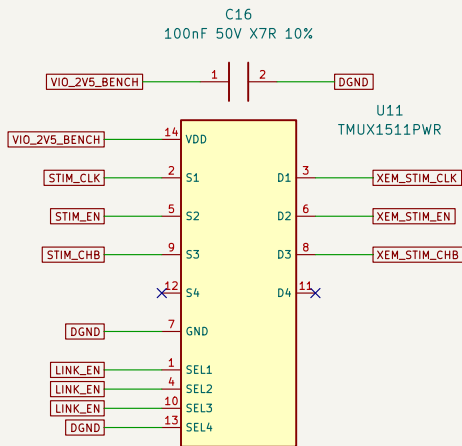
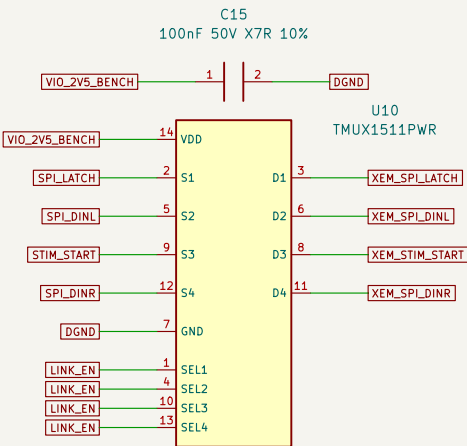
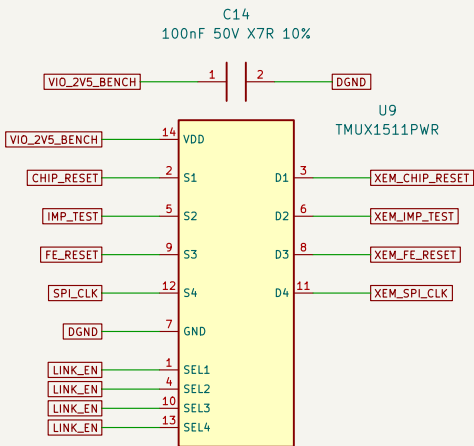
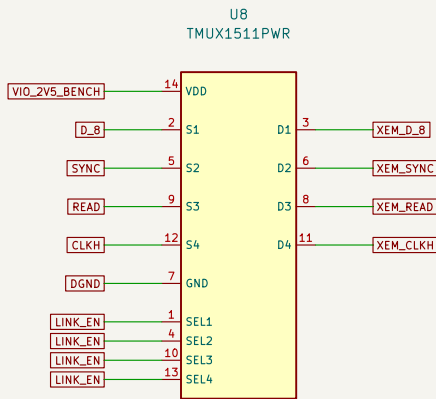
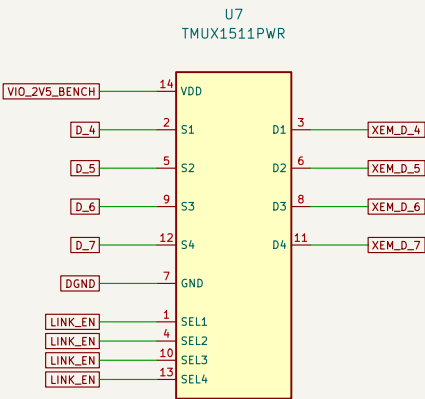
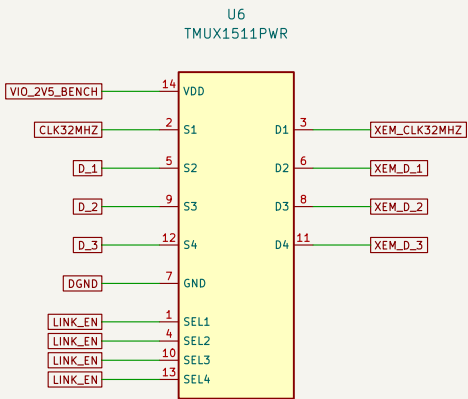
Rev: R3

Size: A5	
KiCad E.D.A. 10.0.6	

Id: 5/8

All 23 original contact assignments are retained through one bidirectional TMUX1511 channel each.

SEL=0 disconnects; LINK_EN requires VCCO and limited3.3V supervision. No level translation.



Bench candidate: startup / timing / mechanics unverified

FPGA PCB project

Sheet: /23 channels: default-off signal isolation/

File: 05_SignalIsolation.kicad_sch

Title: 23 channels: default-off signal isolation

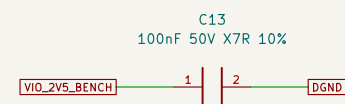
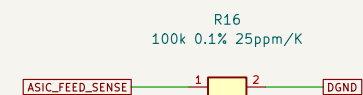
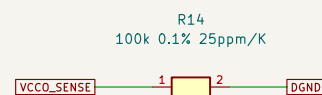
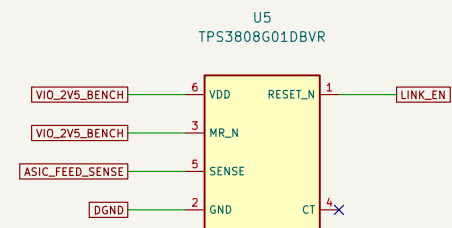
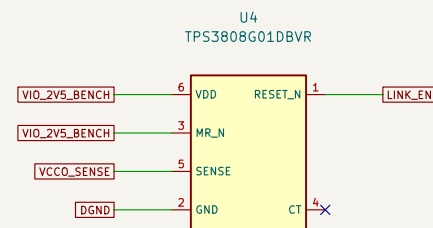
Size: A3

Date: 2026-10-05

Rev: R3

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Id: 6/8



VCC0 assert1.294..1.397V / release<=1.438V. Limited3.3V assert2.794..3.042V / release<=3.133V.

LINK_EN pull-up3.9k/pull-down6.8k from2.5V closes worst24-selector leakage and VIH/VIL margins.

These sense carrier rails. Real ASIC pad limits, rail startup and abrupt-loss behavior require measured acceptance.

Bench candidate: startup / timing / mechanics unverified

FPGA PCB project

Sheet: /Signal-link rail supervision/

File: 06_Link_Supervision.kicad_sch

Title: Independent VCCO and low-stage link supervision

Size: A3

Date: 2026-10-05

Rev: R3

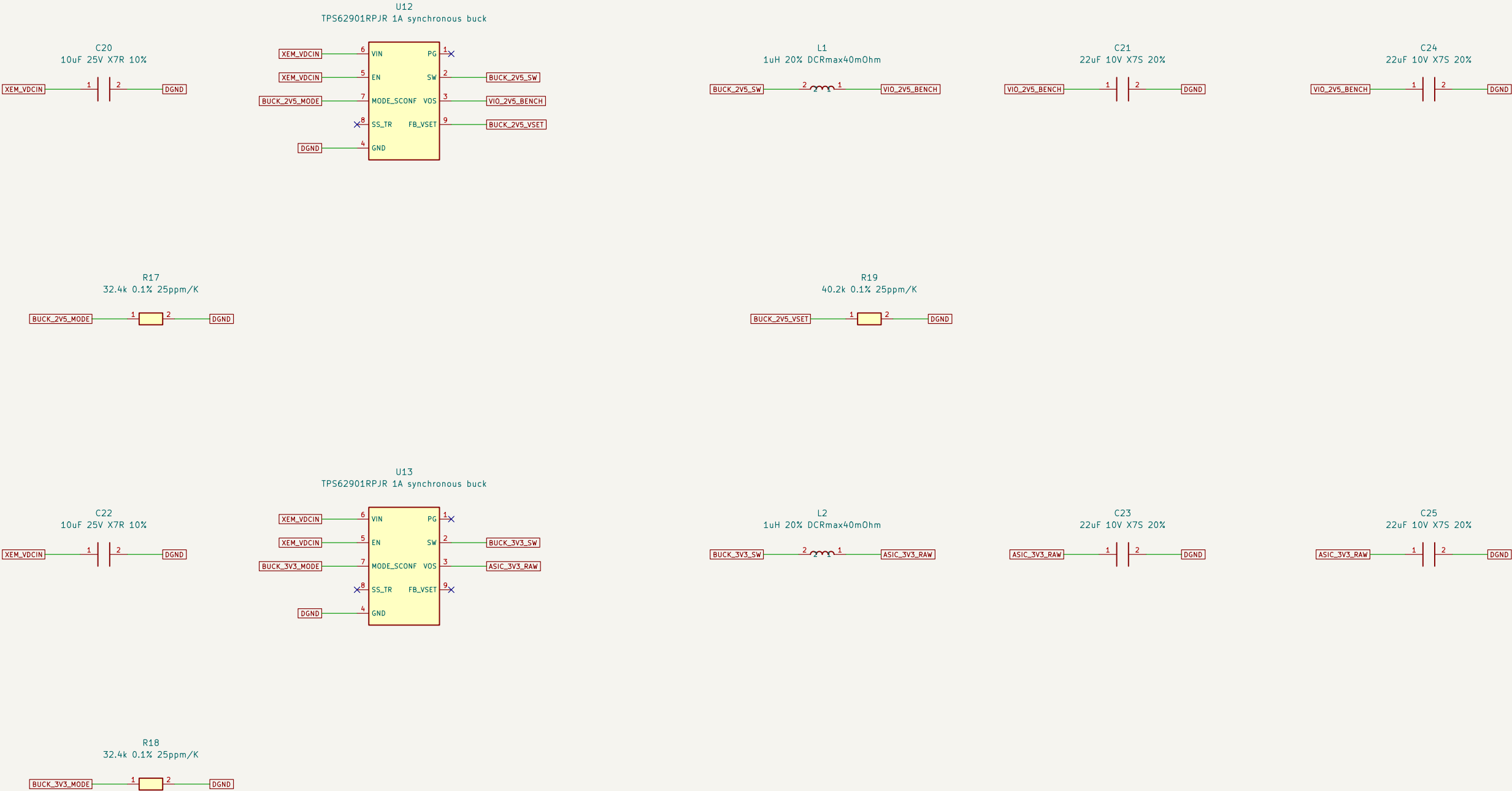
Size: A3	
KiCad E.D.A. 10.0.6	

Id: 7/8

One external12V +/-5% input atJ3. J4/J5 aregeneratedoutputtestpadsonly.

TPS62901 internalVSET: nominal2.5V/3.3V, full-TJaccuracy+/-1.25% atlocaloutputnode.

MODE32.4k selects2.5MHz forcedPWM/internalVSET/active discharge. PG andSS/TR intentionallyNC.



2.5V planning<=160mA includingVCCO/bleeder/controls;3.3V protecteddownstream<=150mA continuous.

Stimulationhigh-voltageinputsandAGNDremainnonoriginalLDO; thiscarrierdoesnotreplace thoseexternal supplies.

Bench candidate: startup / timing / mechanics unverified

FPGA PCB project

Sheet: /12V to2.5V/3.3V converters/

File: 07_Single_12V_Input.kicad_sch

Title: Single external12V input: local2.5V and3.3V bucks

Size: A3

Date: 2026-10-05

Rev: R3

KiCad E.D.A. 10.0.6

Id: 8/8